# Econ 814 Recitation 9 Notes

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## 1 2022 Homework 3 question 2

[1]**Assume that a worker is initially unemployed and that they receive an offer of $w$ drawn from the distribution $F(w)$. Assume that the wage offers are independently and identically distributed across agents. The objective function of the agent is**

$$E\sum_{t=0}^{\infty}\beta^{t}u\left(c_{t},l_{t}\right),\text{ where }\beta\in\left(0,1\right),$$

**where $c_t$ is consumption and $l_t$ is leisure at time $t$. The worker's budget cosntraint is given by**

$$\begin{aligned} c_{t}+a_{t+1} &= \left(1+r\right)a_{t}+q_{t}, \\ q_{t} &= \begin{cases} w_{t}n_{t} & \text{if employed} \\ w^{u} & \text{otherwise} \end{cases} \end{aligned}$$

**where $w_t$ is the wage at time $t$, $a_t$ is the amount of assets owned by the worker at the beginning of period $t$, $r$ is the constant real interest rate, $w^u$ is unemployment compensation, and $n_t$ is labot supply at time $t$.**
**In each period $t$, the worker is endowed with one unit of time that can be allocated across labor and leisure. Assume that $u(.,.)$ has the usual properties. Write down the Bellman equation for this economy.**
The choice for the state variable in this problem is the vector $(a, w, s)$, where $s = E$ if the worker is employed and $s = U$ if the worker is unemployed.
The problem of an employed worker is

$$\begin{aligned} v\left(a,w,E\right) &= \max_{c,l,n,a'} u\left(c,l\right)+\beta v\left(a',w,E\right) \\ &\quad s.t. \\ a' &= \left(1+r\right)a+wn-c \\ l+n &= 1 \end{aligned}$$

[1]Thanks to Prof Jorge Soares for the problems and solutions

The problem of an unemployed worker that receives an offer $w$ is

$$v(a,w,U)=\max_{E,U}\left\{\begin{array}{c} v(a,w,E),\max_{c,a'}\left[u(c,1)+\beta\int v(a',w',U)\,dF(w')\right] \\ s.t.\ a'=(1+r)\,a+w^u-c \end{array}\right\}$$

where the first term is the value of accepting the job, whereas the second term represents the value of remaining unemployed and facing another wage offer next period.

# 2 2022 Homework 3 question 3

**Variable labor supply:**
**An unemployed worker receives each period a wage offer w drawn from the distribution function $F(w)$. The worker has to choose whether to accept the job - and work forever at that wage -or to search for another offer and collect b in unemployment compensation. The worker who decides to accept the job must choose the number of hours to work in each period. The worker chooses a strategy to maximize**

$$E\sum_{t=0}^{\infty}\beta^t u(y_t,l_t),\ \text{where } \beta\in(0,1),$$

**$\mathbf{y_t}$= b if the worker is unemployed, and $\mathbf{y_t}$= w $(1-l_t)$ if the worker is employed and works $(1-l_t)$ hours, where $l_t$ is leisure with $0\leq l_t\leq 1$. Analyze the worker's problem. Argue that the optimal strategy has the reservation wage property. Show that the number of hours worked is the same in every period. Show that the value function is increasing in the wage.**

The choice for the state variable in this problem is the vector $(w,s)$, where $s=E$ if the worker is employed and $s=U$ if the worker is unemployed.
The problem of an employed worker is

$$v(w,E)=\max_{l}\left\{u(\mathbf{w}(1-l),l)+\beta v(w,E)\right\}$$

Then it follows that

$$v(w,E)=\frac{u(w(1-l(w)),l(w))}{1-\beta}$$

where

$$l(w)=\arg\max_{l}u(\mathbf{w}(1-l),l).$$

Given the usual assumptions for the utility function, this problem has a unique solution. Therefore, a worker facing a constant wage of $w$ in each period of their life, will work a constant number of hours equal to $l(w)$.

Let's now show that $v(w, E)$ is increasing in $w$.
Consider $w_1 < w_2$. We have

$$\begin{aligned} u(w_1(1-l(w_1)), l(w_1)) &\leq u(w_2(1-l(w_1)), l(w_1)) \\ &\leq \max_l u(w_2(1-l), l) \\ &= u(w_2(1-l(w_2)), l(w_2)) \end{aligned}$$

Intuitively, a worker receiving $w_2 > w_1$ has the option to work $1 - l(w_1)$ hours paid $w_2$, that yields a higher utility than working $1 - l(w_1)$ hours paid $w_1$. Its optimal choice $1 - l(w_2)$ necessarily yields an even higher utility.

Now consider an unemployed worker. The problem of an unemployed worker is

$$v(w, U) = \max_{E,U} \left\{ v(w, E), u(y, 1) + \beta \int v(w', U) \, dF(w') \right\}$$

The outside maximization is over two actions: accept the offer (in which case the worker chooses $l$ optimally) or reject the offer, collect unemployment compensation, and wait for a new offer next period. The first term is increasing in $w$ and the second is independent of $w$. Therefore the optimal policy is to accept offers that are at least equal to some $\bar{w}$. Once an offer has been accepted, hours worked are constant and equal to $l(w)$.

# 3 Final 22 Q2

(30 points) Consider the following overlapping generations model. In period t there are $N_t$ two-period-lived consumers born, where $N_t = (1+n)^t N_0$, with $N_0$ given and $n > 0$. At $t = 0$ there are some one-period-lived old agents who are collectively endowed with $K_0$ units of capital, and who maximize consumption at $t = 0$. A consumer born in period t has preferences given by

$$u(c_{yt}, c_{ot+1}) = \ln c_{yt} + \ln c_{ot+1}$$

where $c_{yt}$ ($c_{ot}$ ) denotes consumption of a young (old) consumer in period $t$. Each consumer has one unit of time available when young that is allocated to work. The production technology is given by

$$Y_t = K_t^{\alpha} N_t^{1-\alpha},$$

where $Y_t$ is output, $K_t$ is the capital input, $N_t$ is the labor input, and $0 < \alpha < 1$. One unit of the consumption good can be converted into one unit of capital, and vice-versa. Capital fully depreciates after one period. Let $w_t$ denote the real wage rate and $r_t$ the capital rental rate.

## 3.1 Part a

**(5 points) Determine the capital/labor ratio, the consumption of the young, and the consumption of the old in a socially optimal steady state. Assume that the social planner chooses a steady-state that maximizes** $(\log c_y + \log c_o)$.
The resource constraint faced by the social planner is

$$N_t c_{y,t} + N_{t-1} c_{o,t} = (N_t k_t)^{\alpha} N_t^{1-\alpha} + (1-\delta) N_t k_t - N_{t+1} k_{t+1}$$

which, when divided by $N_t$, becomes:

$$\begin{aligned} c_{y,t} + \frac{c_{o,t}}{1+n} &= k_t^{\alpha} + (1-\delta) k_t - (1+n) k_{t+1} \\ &= k_t^{\alpha} - (1+n) k_{t+1} \end{aligned}$$

So, in steady state, the resource constraint faced by the social planner is

$$c_y + \frac{c_o}{1+n} = k^{\alpha} - (1+n)k;$$

where $c_y$ is consumption by the young, $c_o$ is consumption by the old, and $k$ is the capital/labor ratio. The socially optimal steady state allocation is obtained as the solution to the problem of solving

$$\begin{aligned} &\max_{c_y, c_o, k} (\ln c_y + \ln c_o) \\ &subject \;\; to \;\; c_y + \frac{c_o}{1+n} = k^{\alpha} - (1+n)k. \end{aligned}$$

which is equivalent to

$$\max_{c_o, k} \left[ \ln \left( k^{\alpha} - (1+n)k - \frac{c_o}{1+n} \right) + \ln c_o \right]$$

The FOCs are

$$\begin{aligned} k &: \alpha k^{\alpha-1} - (1+n) = 0 \\ c_o &: \frac{-\frac{1}{1+n}}{c_y} + \frac{1}{c_o} = 0 \end{aligned}$$

The optimal capital/labor ratio is given by the solution to

$$\alpha k^{\alpha-1} = (1+n),$$

or

$$k = \left( \frac{\alpha}{1+n} \right)^{\frac{1}{1-\alpha}},$$

note that this is the Golden Rule level of capital, which makes sense as given the objective function the social planner would want to maximize steady-state consumption and then distribute it optimally across old and young. If we define

$$\phi = \left(\frac{\alpha}{1+n}\right)^{\frac{\alpha}{1-\alpha}}(1-\alpha),$$

we can use the optimality condition

$$c_o = c_y(1+n)$$

and the social planner's budget constraint to compute the optimal consumption of the young and the old:

$$c_y = \frac{\phi}{2},\ c_o = \frac{(1+n)\phi}{2}$$

## 3.2 Part b

**(5 points) In period $t$, the government levies a lump-sum tax of $\tau_{yt}$ on each young person, and a lump-sum tax of $\tau_{ot}$ on each old person. Write down the government's budget constraint.**
The government's budget constraint is given by

$$0 = N_t\tau_{yt} + N_{t-1}\tau_{ot}. \quad (2)$$

There is no need to make any assumptions about the sign of the lump-sum taxes at this point.

## 3.3 Part c

**(5 points) Determine the optimal savings of a young consumer given market prices and the taxes described above.**
A young consumer solves[2]

$$\max_{s_t}\{\ln(w_t - s_t - \tau_{yt}) + \ln[(1+r_{t+1})\,s_t - \tau_{ot+1}]\}$$

The FOC condition is

$$-\frac{1}{w_t - s_t - \tau_{yt}} + \frac{(1+r_{t+1})}{(1+r_{t+1})\,s_t - \tau_{ot+1}} = 0$$

and the solution is

$$s_t = \frac{w_t - \tau_{yt}}{2} + \frac{\tau_{ot+1}}{2\,(1+r_{t+1})} \quad (3)$$

[2]Note here I think it should be $r_{t+1}$ rather than $1+r_{t+1}$ below due to complete depreciation

### 3.4 Part d

**(15 points) Suppose that the government introduces a pay-as-you-go social security system whereby transfers are made between the young and the old in each period, so that $\tau_{ot} = -(1+n)\tau_{yt}$; $\tau_{yt} = \tau$ for all t.**

1. **(5 points) Show that the government budget constraint is satisfied in each period.**
   To verify that the government budget constraint is met, substitute

$$\begin{aligned} \tau_{ot} &= -(1+n)\tau, \\ \tau_{yt} &= \tau \end{aligned}$$

   in

$$0 = N_t\tau_{yt} + N_{t-1}\tau_{ot}$$

$$N_t\tau_{yt} + N_{t-1}\tau_{ot} = N_t\tau - N_{t-1}(1+n)\tau = N_t\tau - N_t\tau = 0 \text{ QED} \qquad (1)$$

2. **(5 points) Determine whether or not the government can set $\tau$ so as to achieve the socially optimal steady state in a competitive equilibrium. If the answer is yes, determine the optimal level of $\tau$ and whether the optimal $\tau$ is positive or negative.**
   In equilibrium, we have

$$k_{t+1}(1+n) = s_t, \quad (4)$$

   and from the firm's profit maximization problem,

$$\begin{aligned} w_t &= (1-\alpha)k_t^{\alpha}, \quad (5) \\ r_t &= \alpha k_t^{\alpha-1} - \delta \\ &= \alpha k_t^{\alpha-1} - 1. \quad (6) \end{aligned}$$

   Notice that we exclude the depreciation rate of capital here. We need to take into account the depreciation of capital so either we introduce it here or in the consumers and government budget constraints:

$$\begin{aligned} c_{o,t+1} &= (1 + r_{t+1} - \delta)\, s_t - \tau_{ot+1} \\ 0 &= N_t\tau_{yt} + N_{t-1}\tau_{ot}. \end{aligned}$$

   Substituting using (2), (5) and (6) in (4), we get

$$k_{t+1}(1+n) = \frac{(1-\alpha)k_t^{\alpha}}{2} - \frac{\tau}{2} - \frac{\tau(1+n)}{2\alpha k_{t+1}^{\alpha-1}} \quad (7)$$

   In the optimal steady state, we have

$$k_t = k_{t+1} = k = \left(\frac{\alpha}{1+n}\right)^{\frac{1}{1-\alpha}},$$

and so we can substitute in (7) and solve for the tax $\tau = \tau^*$ that yields this as a competitive equilibrium. We get

$$\left(\frac{\alpha}{1+n}\right)^{\frac{1}{1-\alpha}}(1+n) = \frac{(1-\alpha)\left(\frac{\alpha}{1+n}\right)^{\frac{\alpha}{1-\alpha}}}{2} - \frac{\tau}{2} - \frac{\tau(1+n)}{2\alpha\left(\frac{\alpha}{1+n}\right)^{\frac{\alpha-1}{1-\alpha}}}$$

$\Rightarrow$

$$\left(\frac{\alpha}{1+n}\right)^{\frac{1}{1-\alpha}}(1+n) = \frac{(1-\alpha)\left(\frac{\alpha}{1+n}\right)^{\frac{\alpha}{1-\alpha}}}{2} - \tau$$

$\Rightarrow$

$$\begin{aligned}
\tau^* &= \left(\frac{\alpha}{1+n}\right)^{\frac{1}{1-\alpha}}\left[\frac{(1-\alpha)}{2}\left(\frac{\alpha}{1+n}\right)^{-1} - (1+n)\right] \\
&= \left(\frac{\alpha}{1+n}\right)^{\frac{1}{1-\alpha}}\left[\frac{(1-\alpha)}{2}\frac{1+n}{\alpha} - (1+n)\right] \\
&= (1+n)\left(\frac{\alpha}{1+n}\right)^{\frac{1}{1-\alpha}}\left[\frac{(1-\alpha)}{2\alpha} - 1\right] \\
&= (1+n)\left(\frac{\alpha}{1+n}\right)^{\frac{1}{1-\alpha}}\left[\frac{1-3\alpha}{2\alpha}\right]
\end{aligned}$$

and $\tau^* > 0$ iff $1 - 3\alpha > 0$ which implies that

$$\frac{1}{3} > \alpha$$

Note that when $\tau = 0$ (7) becomes:

$$k_{t+1}(1+n) = \frac{(1-\alpha)k_t^{\alpha}}{2} \quad (7')$$

and the corresponding steady-state level of capital is

$$k = \left[\frac{(1-\alpha)}{2(1+n)}\right]^{\frac{1}{1-\alpha}}$$

and marginal product of capital is

$$\alpha k^{\alpha-1} = \alpha\left[\frac{(1-\alpha)}{2(1+n)}\right]^{-1} = \frac{2(1+n)\alpha}{(1-\alpha)}$$

The marginal product of capital at the social planner's optimum is:

$$1+n$$

3. **(5 points) Explain your results.**
The problem here is that there are no markets for intergenerational trade, but the government can accomplish intergenerational trade through the tax system. The appropriate lump sum transfers between young and old can achieve a socially optimal steady state. However, the optimal transfers could either flow from the young to the old or from the old to the young depending on whether the competitive equilibrium without taxes has a level of capital above the socially optimal, which is the golden rule level of capital, or below. From the above analysis it seems that the transfer is positive when the share of labor in ouptut, $1 - \alpha$, is higher and the share of capital is lower. The reason might be that in those cases young agents have a higher labor income but a low return on capital investment and the only way to attain higher consumption levels upon retirement is to save large amounts of capital, leading to over-acumulation of capital.